

# Lucas A. Saavedra

SOFTWARE ENGINEER · AI RESEARCHER

Buenos Aires, Argentina

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## Summary

I am a software engineer specializing in back-end development, artificial intelligence, computational neuroscience, and machine learning. My experience includes deep learning for neurobiological data analysis and back-end development for cybersecurity applications. Through my skills, I have published in international scientific journals and created open-source software, demonstrating proficiency and innovation in both engineering and science.

## Skills

### Soft Skills

Effective communication, Teamwork, Problem-solving, Critical thinking, Resilience in complex challenges, Stress management and frustration, Fast adaptability

### DevOps

AWS, Docker, CircleCI

### Technologies

Django, REST API, TensorFlow, PyTorch, Pandas, NumPy, SQL, MongoDB, Redis

### Programming

Python, JAVA, C, C++, R, LaTeX, MATLAB

## Education

### Pontificia Universidad Católica Argentina (UCA)

Buenos Aires, Argentina

SOFTWARE ENGINEERING

2018 - 2023

- **Final work:** Comparison of Machine Learning models for the analysis of anomalous diffusion of acetylcholine neurotransmitter receptor trajectories. (Grade: 10)

## Experience

### Deloitte

Buenos Aires, Argentina

BACK-END DEVELOPER

Jan. 2021 - Jun. 2022

- Developed Codex Gigas, a malware "DNA" search engine that identifies patterns and characteristics for malware hunting using REST APIs with Python.
- I worked in Agile environments.
- Built ImageSimilarity, a tool to classify images based on similarity using the metric SSIM.
- Developed applications for ThreatConnect using Python, Docker, MongoDB, and SQL.
- **Technologies used: Python, Docker, MongoDB, SQL, Redis.**

### BIOMED UCA-CONICET, Pontificia Universidad Católica Argentina (UCA)

Buenos Aires, Argentina

RESEARCH ASSISTANT

Jan. 2022 - PRESENT

- Implemented Deep Learning models for static and dynamic analysis of proteins in cells.
- Analyzed spatial distribution and density of proteins using super-resolution microscopy (STED, STORM, etc) and MATLAB.
- Developed an efficient and autonomous GNN-based algorithm in TensorFlow to quantify dynamic particle clustering in molecules data.
- **Technologies used: Python, Java, Tensorflow, Pandas, PyTorch, Docker, NumPy, MongoDB, MATLAB.**

### Facultad de Ingenieria, Pontificia Universidad Católica Argentina (UCA)

Buenos Aires, Argentina

ASSISTANT PROFESSOR

Mar. 2023 - PRESENT

- "Introduction to Data Science": Artificial Intelligence concepts, Neural Networks, Scientific Method.
- "Data-Oriented Programming": Object-Oriented Programming, Python, Java, NumPy, Pandas, UML Software Design.
- "Discrete Mathematics": Number Theory, Graph Theory, Proof Methods, Boolean Algebra.

### UpWork

FREELANCER

Dec. 2023 - PRESENT

- Fixed and hourly contracts.
- Web scraping with Python and Selenium in AWS EC2.
- YOLO training to detect basketball players.
- **Technologies used: Python, YOLO, Pandas, Docker, SQL, EC2 AWS, Selenium.**

## Certifications and Courses

- 2020 **The Data Scientist's Toolbox**, Johns Hopkins University, Coursera
- 2020 **Neural Networks and Deep Learning**, DeepLearning.AI, Coursera
- 2021 **Machine Learning**, Stanford Online, Coursera
- 2021 **Programming Foundations: Design Patterns (2013)**, Linkedin Learning
- 2021 **R Programming**, Johns Hopkins University, Coursera
- 2022 **Natural Language Processing in TensorFlow**, DeepLearning.AI, Coursera
- 2022 **DeepLearning.AI Tensorflow Developer**, DeepLearning.AI, Coursera
- 2024 **Deep Neural Networks with PyTorch**, IBM, Coursera
- 2025 **Introduction to Statistics**, Stanford Online, Coursera

## Publications and conferences

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- 2022 **Mapping the nicotinic acetylcholine receptor nanocluster topography at the cell membrane with sted and storm nanoscopies**, International Journal of Molecular Sciences
- 2023 **The individual molecule and its tribe, the nanocluster: The case of the nicotinic acetylcholine receptor**, 11th International Weber Symposium *Punta del Este, Uruguay*
- 2024 **A supervised graph-based deep learning algorithm to detect and quantify clustered particles**, Nanoscale
- 2024 **Machine learning empowers static and dynamics clustering characterization of transmembrane receptors**, Building Bridges in Computational Biophysics (BBCB) v3.0
- 2025 **Temporal convolutional networks work as general feature extractors for single-particle diffusion analysis**, Journal of Physics: Photonics